TABLE I: Summary of Related Work on FAIR Principles and FAIR-a Tools

Reference or Citation	Contribution	Research Gap
Michaelis, Lea (2023) [28]	The study evaluates FAIR principles' application in the German Network University Medicine (NUM) during the COVID-19 pandemic using tailored assessment tools.	The work addresses the lack of comprehensive evaluations of FAIR compliance and best practices in large research networks like NUM.
Trifan, Alina (2020) [30]	Overview of FAIR principles' impact and evaluation of biomedical platforms using FAIR metrics.	Highlights the novelty and challenges in translating FAIR principles into practice, particularly in assessing FAIR compatibility in biomedical platforms.
Trifan, Alina (2019) [31]	Evaluates FAIR adoption in biomedical platforms using FAIR metrics.	Limited examples exist on translating FAIR principles into practice for biomedical platforms.
Cuno, Alvaro (2020) [34]	Evaluates five stress detection datasets using FAIR metrics and highlights compliance gaps, particularly in interoperability.	Lack of interoperability and FAIRness improvements in stress datasets.
Bahlo, Christiane (2024) [35]	Developed the AgReFed FAIR tool to enhance agricultural data FAIRness.	Existing FAIR tools lack functionality tailored to agricultural research data needs.
Gao, Ruoyuan Jones, Sarah (2020) [37]	Provides an overview of data management planning tools and their role in supporting FAIR principles.	Limited focus on integrating DMPs with FAIR principles and user competency in tool utilization.
Candela, Leonardo (2024) [38]	Analyzes 20 FAIR tools and 1,180 metrics, identifying trends, gaps, and discrepancies in FAIR assessments.	Challenges in aligning metric intents with FAIR principles and addressing recurring issues in assessment tools.
Weber, Tobias (2018) [39]	Proposes a use-case-centric metric to assess FAIR compliance in research data repositories.	Lack of targeted metrics for specific use cases like spatially and temporally annotated images.
Clarke, Daniel J.B. (2019) [40]	Developed the FAIRshake toolkit for community-driven FAIR metrics and assessments of biomedical digital resources.	Limited tools integrating manual and automated FAIR assessments with visualization capabilities.
Gaignard, Alban (2023) [41]	Developed FAIR-Checker for automated FAIRness assessment and metadata improvement.	Automation challenges in FAIRness assessment requiring technical expertise.
Shiferaw, Kirubel Biruk (2024) [44]	Introduced a framework aligning reporting guidelines with FAIR principles to enhance reproducibility and open science.	Lack of integration between FAIR principles and traditional reporting standards to address reproducibility challenges.
Craig, Adam (2019) [45]	Introduced the DREAM principles as a historical precedent to the FAIR principles and defined quantitative metrics for their evaluation.	Questions the originality and accuracy of FAIR principles and metrics, advocating for consistent terminology and fair attribution.
Wilkinson, Mark D. (2023) [46]	Defines FAIR stakeholder categories and explores governance models for FAIR implementation.	No standardization or consistent measurement of FAIR behaviours.
M¨uller, Heimo (2023) [47]	Developed BIBBOX, a platform to help researchers publish datasets and software in a FAIR manner.	Lack of clarity in implementing FAIR principles hinders data sharing in life sciences.
Lien-Talks, Alphaeus (2024) [48]	Highlights the importance of FAIR principles in bioarchaeology and proposes standardised data management and federated search interfaces.	Lack of standardised procedures and limited metadata/documentation impede effective data reuse in bioarchaeology.
Feijo´o, Matheus Pedra Puime (2020) [49]	Introduces the Bio FAIR Evaluator Framework to assess genomic databases' FAIRness from human and machine perspectives.	Existing works lack detailed evaluation of genomic databases from both human and machine perspectives.
Krans, N.A. (2022) [17]	Reviewed ten FAIR assessment tools, categorized their usability, and provided recommendations for users and developers.	Inconsistent implementation of FAIR principles across tools leads to varied assessment outcomes.
Thompson, Mark (2020) [50]	Explores emerging FAIR tools and technologies in their creolization phase, paving the way for practical FAIR implementation.	Lack of mature tools and standards hinders efficient adoption and implementation of FAIR principles.